POSTCAMP EVALUATION

TEN LESSONS LEARNED

This section summarizes key lessons learned from the initial implementation of the IT camps and translates these observations into practical recommendations for future offerings. The lessons are derived from pilot testing, live camp delivery, participant interactions, and post-camp reflection, with particular attention to instructional design, logistics, and resource constraints. Collectively, these insights highlight common challenges and effective strategies relevant to educators operating in single-instructor or resource-constrained environments.

Lesson 1: Mismatch Between Teaching Style and Audience Level

One week prior to the IT camps, a three-day pilot session (one hour per day) was conducted with a fifth-grade volunteer to evaluate the IoT workshop content and instructional approach. During the first session, the participant became disengaged after approximately 30 minutes, an outcome likely attributable to the use of a college-level, lecture-driven teaching style. Similar patterns have been reported in prior research, which suggests that younger learners typically have shorter attention spans and benefit less from extended lecture-based instruction (Yilmaz et al., 2010). As a result, the pilot session was discontinued early to reassess instructional design.

In response to this outcome, the teaching approach, examples, and presentation materials were redesigned overnight to follow a simplified, step-by-step, hands-on tutorial format that emphasized active participation and frequent transitions. When the revised approach was implemented the following day, the student demonstrated sustained engagement throughout the session and successfully completed all planned activities. This experience directly informed instructional revisions adopted for subsequent IT camps and resulted in noticeably improved engagement among younger participants. One exception was Workshop 4 (Webpage Design), where content complexity continued to present challenges, indicating that instructional style alone may not fully compensate for cognitively demanding material at lower grade levels. This lesson underscores the importance of aligning instructional strategies with learner developmental stages, particularly in outreach programs serving K–12 audiences. It also highlights the value of pilot testing as a low-cost mechanism for identifying misalignments between teaching style and audience needs before full program deployment.

Lesson 2: Workshop Content Too Advanced for Younger Participants

Workshop 4 (Webpage Design) was found to be developmentally inappropriate for younger participants, particularly fifth-grade students. The workshop required learners to hand-type HTML code, which introduced cognitive and technical demands that exceeded their current skill levels. Even with scaffolded instruction and copy-and-paste guidance, many students struggled to understand syntax structure, debugging errors, and the abstract nature of text-based markup languages. This difficulty limited meaningful engagement and reduced opportunities for successful task completion.

These observations are consistent with prior guidance indicating that text-based web development environments may not be suitable for younger learners without sufficient prior

exposure or typing proficiency (Code.org, n.d.). While visual and block-based tools used in earlier workshops supported exploratory learning and rapid feedback, the transition to hand-coded HTML represented a steep increase in complexity. As a result, instructional pacing slowed, and student frustration increased.

Based on this experience, it is recommended that Workshop 4 be restricted to participants at least 12 years old or at the seventh-grade level and above. Alternatively, the workshop could be redesigned using beginner-friendly web development tools or visual editors that reduce syntactic overhead. This lesson highlights the importance of aligning content difficulty not only with instructional goals but also with learners' developmental readiness, particularly in short-duration outreach programs where limited time restricts opportunities for remediation.

Lesson 3: Managing Over-Engagement in Hands-On Activities

For students in grades 5 to 8, certain hands-on activities generated extremely high levels of enthusiasm among participants. For example, after implementing the "Pop the Balloon" game, students repeatedly replayed the activity and were reluctant to transition to the next task. A similar situation occurred with the "Don't Shake Me" game, which integrates Raspberry Pi, Sense HAT, AWS text-to-speech services, leading students to continue interacting with the system beyond the allocated time.

While such engagement is desirable, it also created challenges in maintaining the planned schedule and classroom control. This observation highlights the importance of incorporating structured transition strategies, such as time limits, clear stopping cues, or scheduled breaks, when designing highly engaging activities for younger children. However, this over-engagement situation didn't occur in Workshop 3, which included high school and college students, a parent, and two faculty members from two different disciplines. This difference may be attributed to participant maturity levels.

Lesson 4: Incorrect T-Shirt Sizing

An unanticipated logistical issue arose when the "Children's Small" T-shirts supplied by the manufacturer were significantly smaller than expected. As a result, several participants were initially unable to wear the shirts comfortably, leading to minor inconvenience during distribution. The situation was mitigated by ordering additional shirts in a wider range of sizes, allowing participants to exchange as needed.

This experience underscores the value of ordering surplus materials and size variations when working with K–12 populations, particularly in camps with limited flexibility for replacements. While minor, such logistical details can influence participant satisfaction and should be incorporated into future planning checklists.

Lesson 5: Overestimated Time Allocation

Despite careful curriculum preparation, there was insufficient time to complete a planned Python tutorial during Workshop 3. Although materials had been developed and aligned with instructional goals, pacing constraints and earlier activities consumed more time than anticipated. This outcome illustrates the common tendency to overestimate the amount of content that can be covered within short, fixed instructional sessions.

This lesson reinforces the importance of conservative time planning, especially in single-instructor settings where transitions, troubleshooting, and varying learner paces must be managed simultaneously. Future camps benefited from prioritizing core activities, identifying optional extensions in advance, and treating advanced content as enrichment rather than required material. Such adjustments helped ensure that instructional goals were met without compromising engagement or instructional flow.

Lesson 6: Hardware Availability Challenges

Hardware procurement presented a significant challenge due to a global chip shortage that limited the availability of Raspberry Pi 4 devices. Securing a sufficient number of units required extensive coordination, delayed preparation timelines, and increased stress during the planning phase. This experience revealed a critical vulnerability in hardware-dependent instructional models, particularly in outreach programs operating without dedicated procurement support. As a result, early procurement planning and contingency strategies—such as device sharing, reuse across sessions, or identifying alternative platforms—were incorporated into subsequent camp implementations.

This lesson emphasizes the importance of anticipating supply chain disruptions and building flexibility into instructional design. These early challenges informed later improvements in resource planning and contributed to more resilient camp organization in subsequent years.

Lesson 7: Support, Even Minimal, Makes a Difference

Although the IT camps were intentionally designed to operate under a single-instructor model, even limited assistance had a measurable impact on instructional effectiveness. Student helpers and volunteers supported tasks such as troubleshooting, distributing materials, and assisting participants during hands-on activities. This support allowed the instructor to focus more fully on teaching and responding to learner needs. The presence of minimal support improved instructional efficiency, reduced cognitive load on the instructor, and enhanced the overall camp experience for participants.

This lesson suggests that while a “team of one” model is feasible, incorporating even modest assistance—when available—can significantly enhance program quality. For institutions considering similar outreach efforts, planning for optional or intermittent support may offer substantial benefits without undermining the scalability of the single-instructor approach.

Lesson 8: Logistics Can Be as Important as Instruction

As used in this paper, “logistics” refers to the practical arrangements required to run the camps, including room setup, equipment preparation, software installation, scheduling, and participant coordination. Seemingly minor logistical details—such as room setup, device availability, and equipment compatibility—had a major impact on the overall experience. Smooth logistics allowed the instructor to focus on teaching, while logistical issues could quickly disrupt the flow of the camps. Several unexpected logistical challenges were identified. For example, although all software and hardware were tested in advance using office computers, two weeks prior to the IT camps, it was discovered that the existing computer lab monitors only supported VGA or DVI connections, while Raspberry Pi 4 devices required HDMI connections.

This required a special request to the IT Help Desk to temporarily replace all monitors. In addition, the Raspberry Pi 4 does not have a built-in speaker, so USB soundbars were also requested to enhance the learning experience for activities such as game design projects (e.g., the Pop the Balloon game, Don't Shake Me game, multimedia animation activities).

These experiences underscore the importance of verifying the full instructional environment—including peripheral compatibility and classroom infrastructure—well in advance of camp delivery, particularly in hardware-intensive settings.

Lesson 9: Identifying Sustainable Post-Camp Learning Opportunities for Students

A key question following the camps was how participants could continue learning after the program concluded. Since most of the hardware (e.g., Raspberry Pi, Sense HAT, sensors, and other IoT components) belonged to the university, it was not feasible to allow ongoing access for students and parents/guardians.

One potential solution is collaboration with local public libraries. Libraries may allocate small budgets to purchase similar devices and allow students to borrow them. Such partnerships could support informal, self-directed exploration while leveraging existing community infrastructure. This approach suggests a scalable and sustainable model for extending computing education opportunities in rural and underserved areas, while reinforcing community–institution partnerships.

Lesson 10: Informal Feedback Is Still Valuable

Although formal survey instruments were not implemented due to inexperience during the initial camp offering, informal feedback from participants and parents or guardians proved to be a valuable source of insight. Observations of student behavior during activities, questions posed throughout sessions, and post-camp conversations all contributed to identifying strengths and areas for improvement in instructional design and logistics. These informal data complemented the qualitative evaluation of learning artifacts and engagement patterns described earlier. Together, they informed iterative refinements to curriculum structure, pacing, and activity design in subsequent camps.

This lesson highlights that, while formal assessment methods are desirable, carefully observed informal feedback can still play a meaningful role in guiding continuous improvement—particularly in exploratory, short-duration outreach programs.

Summary of Lessons Learned

Overall, these lessons highlight that successful IT camps do not necessarily require extensive resources, but rather careful planning, adaptability, and a focus on participant engagement. Even when operated by a single instructor, a well-structured approach can lead to meaningful, scalable, and sustainable learning experiences. Table 5 (Appendix) summarizes the key lessons learned and corresponding practical recommendations derived from the implementation of the IT camps.

The key lessons in Table 5 show that teaching methods and curriculum complexity must be age-appropriate, as using college-level instructional styles or overly advanced material led to student disengagement and learning difficulties. Hands-on, step-by-step, and interactive activities

were critical for maintaining attention and improving engagement among younger learners. Instructional time was often overestimated, highlighting the need to narrow scope and prioritize essential activities.

Logistical challenges, including hardware incompatibility, insufficient technical preparation, and equipment shortages, caused disruptions and underscored the importance of early testing, advance planning, and contingency strategies. While operating as a single instructor increased workload, even limited assistance from student helpers significantly improved instructional flow and efficiency. Finally, limited access to equipment after the camp constrained continued learning, suggesting that partnerships with local libraries can support longer-term sustainability.